

SOC Phishing Email Investigation Report

Portfolio Lab | Simulated Evidence | Analyst: Abdul Rahman Sesay

1. Executive Summary

The SOC investigated a simulated email impersonating Microsoft Account Support. The message used urgency, a look-alike sender domain, a credential-verification lure and an unexpected ZIP attachment. Simulated SPF, DKIM and DMARC results failed. Based on the combined evidence, the lab verdict is Malicious with High severity.

2. Scope and Safety

This project is designed for portfolio and training use. Publishable indicators use reserved documentation values. No live malicious attachment is included, and no real organisation or user was targeted.

3. Initial Triage

Observed red flags included brand impersonation, urgency, a verification request, a look-alike sender domain and an unexpected archive attachment.

4. Header Analysis

The From, Reply-To, Return-Path, Received and Message-ID fields were reviewed. Authentication-Results in the simulated header show SPF=fail, DKIM=fail and DMARC=fail. These results increase suspicion when combined with the impersonation indicators.

5. URL and Attachment Analysis

The URL was extracted without visiting it directly. The domain uses a simulated look-alike spelling. The ZIP attachment is treated as suspicious and would be hashed and analysed only in an isolated environment. This package intentionally contains no malware.

6. Verdict and Severity

Verdict: MALICIOUS (SIMULATED). Severity: HIGH. Threat type: credential phishing with potential malware delivery. The verdict is supported by impersonation, authentication failures, a credential lure and an unexpected attachment.

7. Response Recommendations

Quarantine the email; hunt for matching messages and IOCs; block confirmed malicious indicators; identify user interaction; reset credentials/revoke sessions if compromise is confirmed; investigate endpoints if execution occurred; and document/tune detections.

8. Indicators of Compromise

Type	Indicator	Assessment
Sender	support@micr0soft-security.example	Simulated look-alike sender
Reply-To	verification@micr0soft-security.example	Simulated suspicious reply address
Domain	micr0soft-security.example	Simulated typosquatting pattern
URL	https://micr0soft-security.example/verify	Simulated credential-phishing URL

Type	Indicator	Assessment
IP	192.0.2.76	TEST-NET-1 documentation IP
Attachment	Account_Update.zip	Simulated unexpected archive
SHA-256	LAB-SIMULATED-HASH-NOT-A-REAL-FILE	Placeholder only; no malware supplied

9. MITRE ATT&CK; Mapping

T1566 - Phishing; T1566.002 - Spearphishing Link; T1566.001 - Spearphishing Attachment. Mappings are limited to behaviours supported by the simulated evidence.

10. Visual Walkthrough

Project 2: Phishing Email Analysis & Incident Response Lab

A complete SOC investigation of a real-world style phishing email, including header analysis, URL and attachment analysis, IOC extraction, MITRE ATT&CK mapping, and remediation recommendations. This project demonstrates practical SOC L1/L2 skills for a portfolio or CV.

1. Suspicious Email Received

Action Required: Update Your Account Information

To: Microsoft Support <[support@microsoft-security.com](#)>

Account_Update.zip
245 KB

Microsoft

Hello,
Your Microsoft account will be suspended due to unusual activity. Please verify your account information to avoid interruption.

[Verify My Account](#)

This is an automated message. Do not reply.

2. Email Header Analysis

Return-Path: <[support@microsoft-security.com](#)>

Received: from mail.microsoft-security.com (185.224.139.76) by mail.company.com with ESMTP id abc123; Mon, 12 Feb 2026 10:14:12 +0000

From: "Microsoft Support" <[support@microsoft-security.com](#)>

To: user@company.com

Subject: Action Required: Update Your Account Information

Message-ID: <2026021210142.12345@microsoft-security.com>

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="-----NextPart_000_123"

3. Sender & Domain Analysis

Domain: microsoft-security.com

Registrar: NameCheap, Inc.

Registered On: 2026-01-15 (Recently registered)

Expires On: 2027-01-15

Updated On: 2026-01-15

Status: clientTransferProhibited

Name Servers: dns1.namecheap.com, dns2.namecheap.com

4. URL Analysis

Hovering over the link shows a different URL:
[https://microsoft-security.com/verify](#)

VIRUSTOTAL

23 security vendors flagged this URL as malicious

URL	Status	Content type	Last analysis
https://microsoft-security.com/verify	Malicious	text/html	2026-02-12 11:05:32 UTC

5. Attachment Analysis

Account_Update.zip
245 KB

VIRUSTOTAL

48 security vendors detected this file as malicious

Type of file	Size	Modified	File hash	File type	Detection	Size	Last analysis
Compressed (zipped) Folder (.zip)	245 KB (250.534 bytes)	12 February 2026, 10:13:41	f3b6d2be0c9f9d7a1e2b3c4d5e6f7ab8dc01e23a8c5d6f7ab01c2d3e	Win32/CIP	Trojan.Generic	245 KB	2026-02-12 11:10:27 UTC

6. IP Address Analysis

185.224.139.76

Summary WHOIS Reputation Malware Passive DNS

IP Address: 185.224.139.76

Country: Russia

Organisation: Cloud provider (VPS)

Reputation: Malicious

Abuse reports: High

Last seen: 2026-02-12

Related domains: microsoft-security.com, update-account-login.com

7. Indicators of Compromise (IOCs)

Type	Indicator	Description
Sender Email	support@microsoft-security.com	Phishing sender address
Domain	microsoft-security.com	Malicious domain (typosquatting)
URL	https://microsoft-security.com/verify	Phishing URL
IP Address	185.224.139.76	Malicious sending IP
Attachment	Account_Update.zip	Malicious attachment
File Hash (SHA256)	f3b6d2be0c9f9d7a1e2b3c4d5e6f7ab8b9cd01e23a8c5d6f7ab01c2d3e	Detected as Trojan.Generic

8. MITRE ATT&CK Mapping

Technique ID	Technique	Evidence from Email
T1566	Phishing	Email attempts to trick users into providing credentials
T1566.002	Spearphishing Link	Malicious URL to a fake Microsoft login page
T1566.001	Spearphishing Attachment	Malicious ZIP attachment containing a Trojan
T1204	User Execution	Victim may execute the attachment
T1071	Application Layer Protocol	Uses standard HTTP/HTTPS for C2 / phishing page

9. Verdict & Severity

Verdict: MALICIOUS

Severity: HIGH

Threat Type: Phishing (Credential Theft & Malware Delivery)

Likelihood: High

Potential Impact:

- Credential compromise
- Malware infection
- Unauthorised access to corporate systems
- Financial and reputational damage

10. Response & Remediation

Recommended Actions

- Quarantine and remove the email from all mailboxes
- Block the sender, domain, URL, IP and file hash
- Investigate if any users clicked the link or opened the attachment
- Reset credentials for affected users
- Monitor for signs of compromise (Splunk / SIEM)
- Conduct user awareness training
- Document the incident and update detection rules

Illustrative workflow: visual reputation counts/details are examples, not live findings.